

Ultra-broadband bidirectional spectrometer for parallel detection

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Abstract: We present a bidirectional reconstructive spectrometer design employing a dual-input-dual-output strategy. This design enables parallel detection without external multiplexers and experimentally achieves a 1000–1750 nm operational bandwidth, paving the way for fully integrated parallel sensing. © 2025 The Author(s)

1. Introduction

Miniaturized spectrometers, particularly those employing reconstructive architectures, have revolutionized spectroscopic analysis by delivering high resolution and broad bandwidth within exceptionally compact footprints, leveraging reconfigurable photonic circuits and a global sampling strategy [1]. Despite these transformative advantages, the single-input-single-output configuration of conventional reconstructive spectrometers imposes a fundamental constraint for parallel detection. Scenarios requiring simultaneous monitoring of multiple light sources or spectral bands—such as hyperspectral imaging or multi-channel process monitoring—necessitate external multiplexing techniques, inevitably impacting system complexity and stability [2].

In this paper, we present a design for a ultra-broadband reconstructive spectrometer (RS) consisting of cascaded Mach–Zehnder interferometers (MZIs) and dispersion-engineered curved directional couplers (DCs), operating on a dual-input-dual-output strategy. The MZI parameters are meticulously optimized for analysing substances with smooth spectral features, e.g., biological tissues and liquids. Integrated thermo-optic (TO) phase shifters facilitate tunable phase combinations, enabling an exponential expansion of sampling channels. Based on the silicon (Si) integration platform, we implement two on-chip reconstructive spectrometers, each incorporating four cascaded MZI stages. Both demonstrate an operational bandwidth of >380 nm, covering the ranges from 1000 nm to 1380 nm and 1330 to 1750 nm, respectively. By co-packaging these chiplets with dedicated bandpass filters, the system seamlessly covers the key NIR detection band from 1000 nm to 1750 nm, essential for tracking organic functional groups such as -OH, -CH, and -CH₂ [3, 4]. This integrated design paves the way for a more streamlined, highly integrated system that enables parallel detection without external multiplexing components.

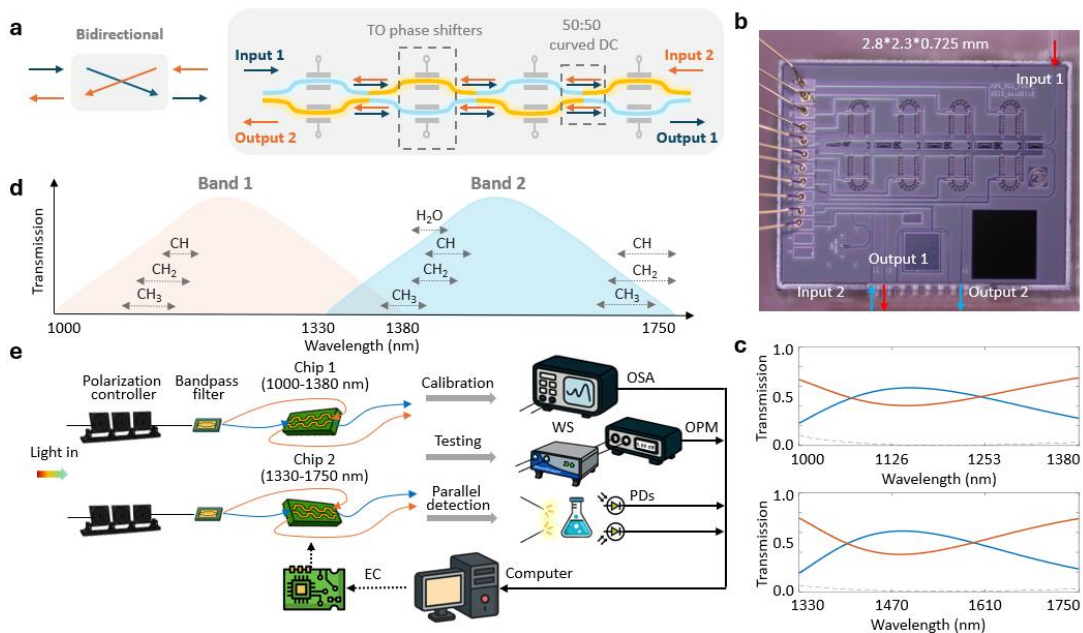


Fig. 1: (a) The structural schematic and internal coupling paths of the proposed bidirectional RS. (b) Micrograph of the fabricated RS. (c) Simulated coupling efficiency of the curved DCs spanning 1000–1380 nm and 1330–1750 nm ranges. (d) Characteristic absorption bands of key functional groups located within the operational bandwidth of the co-packaged spectrometer. (e) Experimental setup for calibration, spectra reconstruction, and parallel detection. EC: Electrical control. OSA: optical spectrum analyzer. OPM: optical power meter. PDs: photodetectors. WS: waveshaper.

2. Principle and characterization

Mathematically, the operation of a reconstructive spectrometer (RS) can be modelled as a matrix multiplication between the incident spectrum ϕ (with N spectral pixels) and a sampling matrix S (with M channels). This operation yields a set of output power intensities that are subsequently detected by a broadband photodetector and used for computational spectral reconstruction. Fig. 1(a) illustrates the principle and Fig. 1(b) shows a microscopic schematic of the RS based on cascaded tunable MZIs. The device occupies a footprint of only 6.44 mm² (2.8 mm × 2.3 mm). To enable bidirectional detection, the device supports dual-port input and output. The symmetric and linear-reciprocal nature of the MZI structure ensures bidirectional functionality through reversible light propagation. The forward- and backward-propagating photons are spatially separated and non-interfering due to the absence of any significant collisional cross-section, enabling their simultaneous transmission without crosstalk. Additional inverse taper waveguides are integrated as internal polarization controllers to mitigate reflection and crosstalk. An arbitrary input spectrum can be sampled by dynamically tuning the MZI responses via TO phase shifters. Our four-stage MZI architecture strikes an optimal balance between spectral resolution, computational load, and hardware cost. By configuring discrete phase shifts between 0 and 2π for each MZI, the total number of sampling channels scales exponentially as the product of all possible phase combinations. Conventional RS designs often employ rapidly fluctuating spectral responses to resolve sharp spectral features by inducing large variations in photodetector output. However, for slowly varying spectra—such as those from human tissues or liquid samples—a gradually varying channel response is preferable to ensure accurate reconstruction. Through finite-difference time-domain (FDTD) simulations in conjunction with particle swarm optimization (PSO), the arm length differences of each MZI were optimized to generate incrementally fluctuated transmission spectra [5]. This approach effectively breaks spectral periodicity and tailors the coupling as well as transmission properties for the two target bands of 1000–1380 nm and 1330–1750 nm. Furthermore, we adopt curved DC architectures, which offers enhanced wavelength insensitivity and phase-matching performance compared to conventional symmetric DCs. As shown in Fig. 1(c), the optimized curved DC achieves a nearly ideal 50:50 coupling ratio across both target wavebands. Finally, the two chiplets are co-packaged with dedicated bandpass filters to form a single, ultra-broadband bidirectional spectrometer operating continuously from 1000 nm to 1750 nm. This integrated approach effectively creates a unified device capable of parallel detection across a key near-infrared range essential for tracking various organic functional groups at distinct wavebands (Fig. 1(d)). For instance, in the detection of glucose solutions, the dual-band operation allows for the concurrent acquisition of the principal absorption peaks of water and glucose, which serves to facilitate cross-validation. Furthermore, the bidirectional synchronous detection scheme not only suppresses experimental noise but also substantially improves the efficiency of transient detection. Fig. 1(e) demonstrates the entire experimental setup for calibration, spectra reconstruction, and parallel detection applications. Data acquisition, and channel selection during the experiments are all coordinated by an integrated computer-electronic control system.

3. Results and Discussion

To evaluate the performance of the RS, we begin by calibrating the sampling matrix with an optical spectrum analyser. Fig. 2(a) displays a selection of experimentally calibrated and channel responses, normalized to eliminate the influence of source spectra and coupling losses, highlighting the strong inter-channel decorrelation observed across distinct spectral windows. Note that the maximum experimentally calibrated wavelength was limited to 1700 nm due to constraints in the available experimental equipment. A series of optical signals were then introduced bidirectionally into the device, including single-peak signals with progressively narrowed linewidths and broadband waveforms generated by a benchtop programmable waveshaper, allowing a comparative analysis of reconstruction capability in dual direction operation. The corresponding output intensity of each channel was recorded using a high-speed optical power meter. The spectral reconstruction was performed via a CVX-based regression algorithm to solve a dedicated optimization function [6]. Fig. 2(b) demonstrates the accurate recovery of narrowing peaks, confirming a maximum achievable spectral resolution better than 2 nm. Reconstruction of broadband signals yielded relative errors between 0.031 and 0.098, as summarized in Fig. 2(c). Evaluation of the experiment results confirms the high accuracy and excellent agreement between the reconstructions obtained from the two opposite directions. Finally, we applied simulation to evaluate the device's practical applicability using the spectra containing common functional groups, such as skin moisture and glucose solution (Fig. 2(d)). The spectrometer successfully resolved different characteristic absorption features and significant concentration differences with high fidelity, indicating its strong potential for real-world scenarios including biological sensing and material analysis. The reconstruction results successfully resolved the position and intensity of characteristic absorption features, indicating the potential of the proposed RS for practical scenarios including biological sensing, material analysis and liquid discrimination.

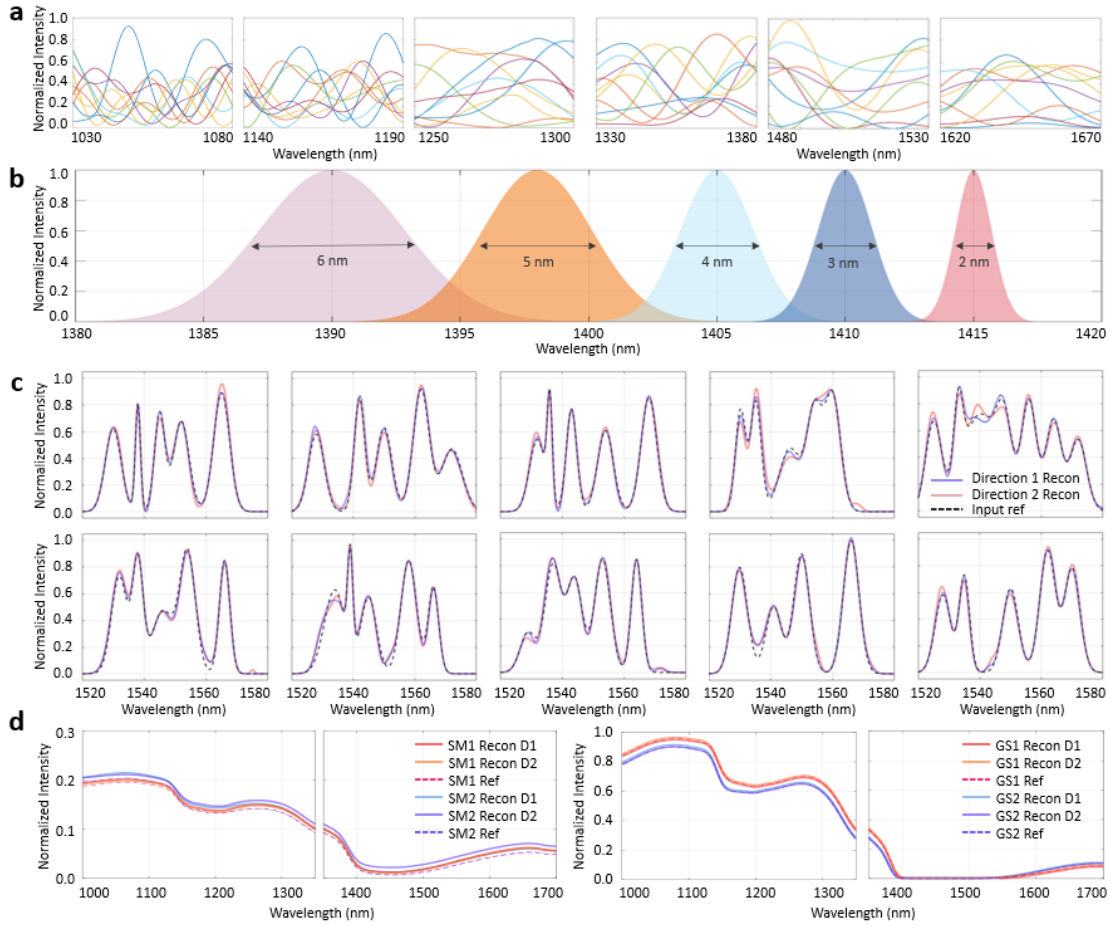


Fig. 2: (a) Measured channel responses in the wavelength ranges of 1000–1380 nm and 1330–1700 nm. Four insets illustrate the channel decorrelation behavior. (b) Reconstructed narrowing peaks with linewidths progressively reduced to 2 nm. (c) Broadband waveshaper spectra reconstructed from two opposing directions: direction 1 (blue) and direction 2 (red). (d) Simulated bidirectional reconstruction results of real-world spectra corresponding to varying concentrations of skin moisture and glucose solution. SM: skin moisture. GS: glucose solution.

4. Conclusion

This work experimentally demonstrates an ultra-broadband bidirectional RS utilizing cascaded MZIs and dispersion-engineered curved DCs. The fabricated chiplets each exhibit an operational bandwidth exceeding 380 nm, collectively covering 1000–1750 nm. The consistent reconstruction fidelity under bidirectional operation verifies the robustness and scalability of the proposed architecture for parallel and multiplexing-free spectral detection. These results establish a promising platform for portable spectroscopic systems targeting precise material characterization and substance identification across broad spectral domains.

5. Acknowledgements

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